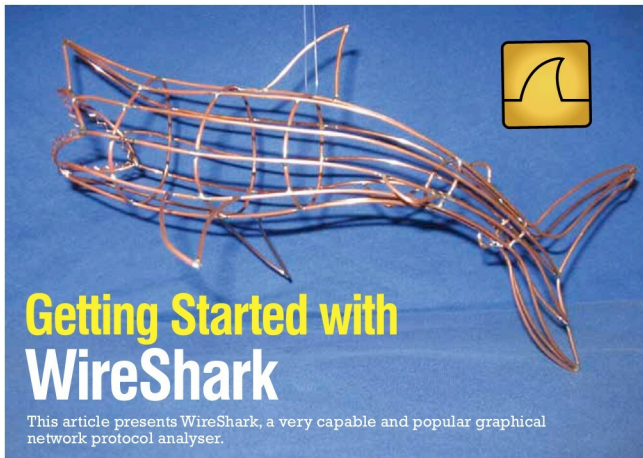




Getting Started with WireShark

This article presents WireShark, a very capable and popular graphical network protocol analyser.

Gerald Combs created *Ethereal*, the ancestor of WireShark, back in 2006. When he changed his job, he could not use the name *Ethereal* any more, so he renamed his software WireShark. Nowadays, most people only know WireShark! The main advantage of WireShark is that it is a graphical application. There is also a command-line version of WireShark but I have never used it.

You can get WireShark either from its website—by compiling its source code—or directly from your Linux distribution. I personally prefer the second option.

Running WireShark

If you try to run WireShark as a normal user, you may not be able to use network interfaces for capturing network traffic, due to reasons related to UNIX permissions. Run WireShark as the root (*sudo wireshark*) when capturing data, and as a normal user when analysing network data. Figure 1 shows WireShark run by a user without root privileges.

Before going into more details about WireShark, I have to talk about network traffic in Ethernet networks that use the TCP/IP family of protocols. When we say TCP/IP, we not only mean the TCP and IP protocols, but many other protocols including ARP, BOOTP, UDP, ICMP, FTP, etc. Information is transferred using packets. Each packet has a header and a body part. The header part contains information that is needed by

the protocol, whereas the body part contains the data. Some protocols are reliable, whereas others are not, which means they do not guarantee package delivery—this is not always a problem, but the application must deal with it, if required.

WireShark captures packets, and analyses and displays them in a human-readable format. WireShark allows you to follow a TCP/IP ‘conversation’ between two machines, view packet data, etc. Before you start capturing, it is better to have in mind a particular issue that you want to solve or examine. What follows are the initial steps towards successful network traffic analysis.

Basic usage

After running WireShark with the required user privileges, you will be able to see the list of available network interfaces. The easiest way to start capturing network packages is to click your preferred interface, which will make WireShark immediately start capturing network packets (Figure 2). If you know nothing about TCP, IP or UDP, you may find the output confusing. To stop capturing, select *Capture* and then *Stop* from the WireShark menu. Alternatively, press the fourth icon from the left, with the white x in a red background, which is only active during captures.

Using the method described for capturing, you cannot change any of the default WireShark capture options (viewable by selecting *Capture* and *Options* from the menu as shown in